

- Q.27 Discuss the working of spray ponds with its neat diagram.
- Q.28 Write the difference between batch dryer and continuous dryer
- Q.29 What are tower packings. Write their uses and properties
- Q.30 Discuss conditions of equilibrium between gas and liquids
- Q.31 Express the relation between film and overall mass transfer coefficient.
- Q.32 Differentiate between humidification and dehumidification with example.
- Q.33 Draw and explain in brief rate of drying curve.
- Q.34 Define loading and channeling.
- Q.35 Discuss any two of the following
- Dry bulb temperature
 - Wet bulb temperature
 - Humid heat

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 What is drying? Explain the construction and working of tray dryer with the help of neat diagram
- Q.37 Differentiate between
- Forced and Eddy diffusion
 - Absorption and Desorption
- Q.38 With the help of neat diagram explain the working principle of natural draft cooling tower

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4th Sem / Chem, P & P Subject:- Mass Transfer - 1

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The film theory of mass transfer assumes:
- Instantaneous equilibrium at the interface
 - The presence of a stagnant film where diffusion occurs
 - No concentration gradient in the film
 - Uniform concentration throughout the bulk
- Q.2 The height of a packed column for gas absorption is determined using:
- Number of Transfer Units of (NTU)
 - Overall Mass Transfer Coefficient
 - Height of Transfer Unit (HTU)
 - All of the above
- Q.3 Which problem is associated with excessive liquid flow in a packed column?
- Flooding
 - Channeling
 - Weeping
 - Loading
- Q.4 The term 'humidity' refers to:
- The temperature of the air
 - The amount of moisture in the air
 - The pressure of the air
 - The volume of the air

- Q.5 The phase of drying where the rate of drying is constant is known as the
 a) Falling rate period b) Constant rate period
 c) Equilibrium period d) Transitional period
- Q.6 The moisture content that is in equilibrium with a given relative humidity of the drying air is called :
 a) Free moisture content
 b) Bound moisture content
 c) Equilibrium moisture content
 d) Critical moisture content
- Q.7 The temperature at which air becomes fully saturated with moisture is called:
 a) Dry bulb temperature
 b) Wet bulb temperature
 c) Dew point temperature
 d) Relative humidity
- Q.8 What is the role of diffusion in mass transfer
 a) To equalize temperature
 b) To mix different phases
 c) To transfer mass from high concentration to low concentration
 d) To reduce pressure differences
- Q.9 Which piece of equipment is commonly used in gas absorption?
 a) Spray dryer b) Tray dryer
 c) Packed tower d) Rotary dryer
- Q.10 What does a humidity chart help to determine?
 a) Dry bulb temperature
 b) Wet bulb temperature
 c) Dew point d) All of the above

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Write names of any two mass transfer operations
 Q.12 Define distillation.
 Q.13 Expand NTU
 Q.14 Write any one use of humidity chart
 Q.15 Write the full form of HETP
 Q.16 Draw any one packing diagram used in packed column
 Q.17 What is dew point
 Q.18 Define drying.
 Q.19 Define Raoult's law
 Q.20 Write the name of any two types of dryers used for drying.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write in brief about mass transfer classifications.
 Q.22 Discuss the role of diffusion in mass transfer. Explain any one type of diffusion.
 Q.23 Explain Fick's law of diffusion with its mechanical relation
 Q.24 Discuss the penetration theory of mass transfer.
 Q.25 Draw and explain any two cooling tower arrangements
 Q.26 Write the condition to be followed while choosing a solvent for absorption.

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